standard deviation, as a proxy of risk, on the x-axis. The idea is simple; we want to achieve as high an annual return as possible at the expense of as low a volatility as we can. The volatility is essentially the currency in which we pay for returns.

Figure 5.13 tells us right away that if we start with a portfolio of only the stock index and keep gradually adding allocation to the futures strategy, we get a lower and lower standard deviation while at the same time the annual compound return rises. The obvious conclusion from a quick inspection of this chart is that it does not make sense to have less than 30–40% futures and the remaining part in equities, because anything less means a lower return at a higher or equal risk level.

Table 5.5 shows the same data as in Figure 5.13 but it also has an additional column with the Sharpe ratio for each point in the curve. Even though 100% allocation to the futures gives the best overall return, the Sharpe ratio of the strategy can be enhanced by adding stocks to the mix.

Naturally, in a real-world scenario, you have more than two assets to choose from and there are other real-life complications as well, which makes it tricky to determine an exact optimal amount to buy of each asset class. Therefore, view this as a guideline approach and a way of demonstrating a concept more than exact numbers.

TABLE 5.5 **Adding diversified futures to an equity portfolio**

Percent futures	Return	Volatility	Sharpe ratio
100	16.16	23.71	0.68
90	15.55	21.10	0.74
80	14.94	18.69	0.80
70	14.33	16.56	0.87
60	13.72	14.84	0.92
50	13.10	13.66	0.96
40	12.49	13.19	0.95
30	11.88	13.50	0.88
20	11.27	14.54	0.78
10	10.65	16.16	0.66
0	10.04	18.22	0.55